Reviewer 1

The manuscript presented by Perez et al. consists of an extensive dataset including 3D reconstructions of oil palm trees, alongside gas exchange measurements (CO2 and H2O) at the whole plant level as well as leaf photosynthesis and temperature measurements. These data were obtained in microcosms under eight environmental conditions, with light, temperature and CO2 as factors.

This extensive dataset will be of great interest to modellers seeking to calibrate or evaluate their models. This is particularly relevant for Functional Structural models (FSPMs), which are difficult to fully parameterise due to the rarity of comprehensive experiments describing both the structure and functioning of plants at the organ level in multiple environments. I was also wondering if the authors could enrich their dataset further by including measurements of organ biomass and biochemical content. The data and the different scripts used in the present study are available online and described in detail at palmstudio.github.io. The documentation is very nice comprehensive and also includes notebooks, allowing the reader to explore the dataset.

Title and abstract

• Does the title clearly reflect the content of the article? *Yes*

• Does the abstract present the main findings of the study? *Yes*

Introduction

• Are the research questions/hypotheses/predictions clearly presented? *Yes*

• Does the introduction build on relevant research in the field? *Yes*

My main comment on the manuscript concerns the definition of its purpose, which is somewhat ambiguous; as the authors did not clearly state whether they intended to present a data paper or a research paper aiming at the parametrisation of a model/digital twin. Assuming that the manuscript falls into the former category, I would encourage the authors to make this clear throughout the manuscript to avoid any confusion. For instance, the chapter "Data for calibrating a digital twin of the experiment could lead the reader to believe that a digital twin was developed in this work. The same applies to page 10 line 5 and page 12 line 2 ("the computation of the digital twin necessitated…").

I also feel that the paragraph on page 3, exposing the advantages of FSPMs, is not very relevant to the present manuscript.

Materials and methods

• Are the methods and analyses sufficiently detailed to allow replication by other researchers? *Yes*

On page 6, the authors state that they 'cloudy conditions' were simulated by reducing the radiation intensity from 300 to 130 µmol m-2 s-1. However, I think it would be more accurate to described this scenario as 'low PAR', since cloudy conditions are mainly characterised by highly diffuse radiations, which does not appear to have been modified in the present experimentation.

Also, could the authors also specify what the temporal dynamics of air temperature, humidity and PAR were used in the different scenarios?

On page 7, some of the colours used in figure 3 are difficult to distinguish (for instance between 600 and 800ppm). It would also be useful to add a table showing the number of replicates for each scenario and plant.

On page 8, could the authors clarify whether the A-Ci and A-PAR curves were obtained by gradually increasing the CO2/PAR or by taking several steps forward and back?

Results

• Are the results described and interpreted correctly? *Yes, if the manuscript is considered as a data paper, the main results are correctly illustrated. However, I was unable to find the supplementary material files.*

On page 7, there are significant daily variations in CO2 in the 400 and 600 ppm. Could the authors comment on these? On page 9, should the x-axis title of figure 4B read 'VPD'?

On Page 10 line 5: the sentence should be in the present tense.

On page 14, it would be helpful to indicate the position of the pots in figure 7A, even though I understand that the plants were not in the chamber during these measurements. In figure 7B, does the grey column correspond to the position of the stipe, which prevented the use of the Sunscan?

On page 15, could the authors mention which climatic scenario was used for figure 8. Positioning the day period on figure 8B would have been interesting too.

On page 17, I was a bit surprised to see that the whole-plant photosynthesis decreased between 600 and 800 ppm (figures 9 and 10). Could the authors comment on this result?

I was also wondering if the photosynthesis units in figure 9 were correct. At maximal irradiance, the whole-plant assimilation would be of 2.5 µmol of CO2 per second at 400 ppm, which is only ~5 µmol m-2 s-1 (according to plant leaf area shown in figure 6). This is also inconsistent with figure 4 and figure 10, which indicates that the daily mean photosynthesis of plants at 400 ppm was 6-8 µmol s-1 (considering 12-hour days from figure 9).

On page 18, I did not understand the origin of the variability in incoming PAR? Were these fluctuations caused by the authors or, did they use the natural variations of light above the microcosms (or from the LED systems) to assess the impact of environmental differences between the LICOR chamber and the rest of the plant? Were these fluctuations regular throughout the whole experiment? Figure 11 shows that the photosynthesis is very low again (same problem as in figure 9?).

Reviewer 2

General comments:

The manuscript by Perez et al. relates to a dataset corresponding to various measurements made on four individual oil palm plants grown in controlled conditions. The authors aim to provide detailed, quantitative data about plant shoot architecture and various biophysical variables at the whole-plant scale or at the scale of the shoot organ in response to various environmental conditions, which FSPM and digital twin approaches may use. The dataset is indeed valuable for this purpose. Here are my main remarks:

1) Structure of the article: The structure of the manuscript is unconventional. In particular, it does not contain any Results, Discussion or Conclusion. However, some results are presented here and there, which is not coherent with a Data paper either. In any case, a discussion about the limits of the experimental setup and approach could - and should - be added [see below].

2) The Abstract and the Introduction are mainly constructed around the idea that this dataset can be used in combination with FSPM. However, no example or methodological approach for doing so is actually presented, and the manuscript corresponds instead to the description of the

experimental method and a few experimental results obtained. If the application to the FSPM is very promising, this should remain a perspective rather than the main angle of this manuscript.

3) The measurements relate to four individual plants only, with no replicate, on which distinct series of different climate scenarios were applied. I understand that producing such a dataset is a huge work, and that relating organ-scale measurements with shoot architecture is not easily replicable from plant to plant anyway. However, different climate scenarios succeeding to each other over time certainly have an impact on plant physiology and leaf gas exchanges. If authors assume that phenomena such as repeated stress conditions or thermo-priming do not play any consequence on the measurements made over time, they should at least state it in the manuscript - and probably discuss it.

4) Nutrient status: No information is provided about the soil substrate used to grow plants, or whether the nutrient needs of the plant was fulfilled at the time of the measurements or not.

5) The example of leaf temperature profiles keeps puzzling me, as plant leaves seem to have reached 50°C in the afternoon - is this really what authors aimed to reproduce in their microcosm?

Minor comments:

Abstract:

“Here, we provided” (here and throughout this paragraph): It is more common to use present time than past time in this situation.

Typo in “Flux measurements”

Rephrase “Such data could be used for future comparisons between simulated and observed physiological behaviours could be used to evaluate the quality of the physiological formalisms independently”

The end of the last sentence is not necessary, i.e. “, which can be applied in various fields, from cognitive studies to decision support applications.”

Introduction:

Rephrase “dynamically varying” - a time variation is necessarily dynamic…

M&M:

Rephrase “We did not intent to have replicates in this experiment, but rather explore”

“(light from 6 am to 6 pm)” - I guess you mean that artificial light was complementing the natural light to reach 600 µmol/m2/s, so it would be useful to precise this

The size of the pots should also be mentioned. What about soil properties?

“The pot of the plant was sealed before entrance into the monitoring microcosm to avoid water loss to the atmosphere” - please mention whether all gas exchanges were avoided with the soil compartment, as this may have important consequences on net CO2 fluxes measured here.

“the baseline conditions were systematically modified” - do you mean for each of the four plants? It needs to be said earlier on.

Fig. 3: What authors mean by “Grey cells” in the caption is unclear. The figure cannot be read in grey levels or for color-blind readers.

p8: “to estimate the Farquhar-von Caemmerer-Berry (FvCB) of” - I guess you mean “to estimate the parameters from the Farquhar-von Caemmerer-Berry (FvcB) model”

p8: One parenthesis is missing in “Kumarathunge et al., 2019)”. Please adjust the parentheses in “from (Dreyer et al., 2001) and (Medlyn et al., 2002)”.

p10 (here and throughout the manuscript): Plant is made of both aerial and belowground parts. Consequently, “three-dimensional reconstructions of the plants” can be claimed only when both shoot and root architecture are considered. As the present work does not investigate root architecture, it would be more correct to write “3D reconstructions of the aerial parts of the plants”.

p10: “To overcome the challenge of distinguishing overlapping leaves in the central region, …” I understand that individual leaf scans were taken in addition to the LiDAR scans of the aerial parts of the plant. If so, this should be better explained, e.g. how the individual scans were taken.

p10: “because the morphology of oil palm leaves does not elongate and expand over time once they are emitted” - do you rather mean “because the shape of one oil palm leaf does not change over time once it has been emitted”?

p11: “Each time a plant was placed in the microcosm to undergo a climate scenario sequence, we selected the LiDAR point cloud that best represented these dates for reconstruction.” Do you mean “For each climate scenario applied in the microcosm”?

p12: “The idea was to provide values of light intensity” è The idea was to measure actual light intensity at multiple points in the space, and compare it with simulated light intensity

Fig. 7B: The titles “BlackFelt_soil” and “105.4 cm from light” can be misleading, please remove them

Fig. 8: I am puzzled by the y-axis of temperature: does it mean all leaves actually reach an internal temperature of 50°C at 4 pm? Is this realistic and sustainable for the plant? If this is not an error, I guess this would imply that plant physiology and therefore photosynthesis is strongly affected by the microcosm… To which climate scenario does this figure correspond?

p16-17: The mixing between Materials & Methods and Results here is unfortunate - please write a proper Results section to present actual data in a clear way

Fig. 9 and elsewhere: please watch out for the correct subscripting of digits in the chemical formula of compounds (CO2 and H2O)

“providing a consistent foundation for evaluating simulation results generated by reproducing a digital twin oft he experiment through a Functional Structural Plant Model (FSPM).” This needs rephrasing and care. So far, I have not understood how an actual digital twin could be built from all these measurements.

Reviewer 3

General Assessment

This manuscript presents the data acquisition protocol developed to calibrate and validate a biophysical FSPM-type model under contrasted climatic scenarios. Using oil palm as a case study, the dataset addresses an important gap, namely the lack of comprehensive, benchmark-quality datasets spanning organ to whole-plant scales for data-demanding functional–structural plant models. The work therefore has strong potential interest for the modelling community, particularly regarding the definition of standardized protocols for feeding and validating plant digital twins.

Overall, the manuscript is clear and well structured. After introducing the scientific context and objectives, the authors describe the experimental conditions and the eight climatic scenarios, followed by the datasets used for model calibration (leaf-level CO_2 and H_2O gas exchanges, chlorophyll content, plant architecture, PAR measurements) and for model evaluation (leaf temperature, whole-plant CO_2 and H_2O gas exchanges).

However, several methodological and conceptual points require clarification in order to maximize the usability of the dataset.

Major points

1. Clarification of the modelling framework

The introduction would benefit from a more explicit description of the targeted FSPM/digital twin framework and of the specific biophysical processes considered. This clarification would help readers better understand the rationale behind the selected measurements and their precise role in model calibration and validation.

2. Justification of the experimental design

While the climatic scenarios are clearly described, the rationale behind (i) the sequence of scenarios (beyond the statement that extreme temperatures were applied at the end) and (ii) the differences in scenarios between plants should be clarified. In particular, could differences in scenario exposure between plants have induced different physiological responses, potentially limiting direct comparisons among individuals? The possible influence of plant “history” on subsequent responses deserves further discussion.

3. Description and timing of measurements for calibration

Additional methodological details are required for photosynthesis measurements :

- references for A–Ci, A–PPFD, and Gs–VPD protocols,
- justification for estimating Farquhar model parameters only from A–Ci curves.

Furthermore, the timing of several measurements should be more clearly justified in relation to their modelling objective:

- Why were photosynthesis measurements performed before and after climate scenarios, but not during them?
- Why were chlorophyll (SPAD) measurements taken at the beginning of the experimental period, and not just before and after leaf gas exchange measurements?
- Weekly LiDAR scans were performed, yet only four dates were retained for analysis. What criteria guided the selection of these four dates, and why were all dates not used?

In addition, the objectives of some measurements would benefit from clearer explanation. For example:

- The statement that whole-plant gas exchange was used to “investigate whether leaf gas exchange is influenced by overall plant conditions” is not sufficiently explicit.
- The purpose of SPAD measurements “to evaluate formalisms that enable the upscaling […]” should be detailed by specifying the type of formalisms considered.

4. Data for evaluation :

Some methodological details are missing:

- How were leaf temperatures derived from thermal images adjusted for air temperature and relative humidity within the chamber?
- In Figure 9, which days are shown? Are they comparable in terms of scenario? Which plant is represented?
- Regarding the impact of plant light environment on leaf-level gas exchange measurements, strong fluctuations in incoming radiation are shown. It is unclear whether these fluctuations were experimentally imposed. Clarification is needed.

Minor points

- Page 6: “Drier and cooler” environments are described as resulting from increasing relative humidity; this should be corrected, as they are created by decreasing relative humidity.

- Page 7: It is stated that “each plant was sequentially placed in the microcosm for one or more days,” but this is not clearly identifiable in Figure 3. This should be clarified. In addition, grey cells mentioned in the legend (indicating storage microcosm periods) are not clearly visible.

- References to supplementary figures are given in the text, but the supplementary material could not be located.